# RevTrace: Preserving Valid Progress across Trusted Feedback Revisions

Anonymous submission
Regular paper

***Abstract*—Trusted feedback can invalidate speculation while leaving computation and delivered progress usable. RevTrace coordinates full-input eligibility, protected selection, and whole-hypothesis consumption. Frozen inputs confirm owned calls; a separate dependency witness governs visible views. Native calls retain original correction even when their own next predictions are wrong. A selected-trace refinement argument connects durable facts and recovery to the protected program. In a predeclared 24-speaker stratum, computation retention alone increases frontier withdrawals from 12 to 17; joint coordination produces zero and adds 26.645 seconds of deadline credit over 714.110 seconds offered. A matched 32-run native comparison preserves original histories: the final protocol delivers 65.822 of 66.340 deadline-credited seconds and passes both whole-run deadlines in six of eight trials, versus none for conservative execution. Fault and sustained-state tests expose recovery, quality, and resource boundaries.**

***Index Terms*—dependable inference, feedback state, incremental computation, speculative execution, streaming ASR**

## I. Introduction

Feedback-driven speech recognition uses generated text as input to later inference. A trusted correction can change that input, invalidating some speculative work while leaving other work usable. A dependable service must distinguish these cases and preserve useful delivered progress while the trusted path catches up.

Computation retention alone can exacerbate this mismatch. In our predeclared Qwen evaluation on three new eight-speaker cohorts, retaining valid computation improved timely delivery but increased observed audio-frontier withdrawals from 12 to 17 when the Consumer still installed snapshots. Retaining the computation and its compatible whole view together produced zero frontier withdrawals in that stratum. The issue is how valid work becomes sustained service across a trusted update.

One actual two-second interval illustrates the loss. Its complete hypothesis reached the Consumer at 1.444 s, meeting the two-second display deadline. At 1.544 s, a new certified root replaced it with a shorter snapshot, although the whole hypothesis retained a valid dependency path. It remained absent at the original deadline and returned with identical text at 3.720 s through certified delivery, within the eight-second deadline. Both first-delivery obligations were met, yet 37 newly displayed characters, including spaces, disappeared for 2.176 s. This is lost interval coverage under a compatible update, independently of transcription correctness.

Incremental dialogue distinguishes active and committed information and propagates revocation [1]; incremental computation reuses dependencies and stable identities [2]–[4]. SSBD combines old draft reuse with display-only masking in re-translation [5], and PredGen speculates on downstream responses during evolving speech [6]. RevTrace applies these ideas to a protected task whose original selection recursively supplies later ASR inputs. Separate sufficient guards determine whether old computation remains eligible and whether installing a root preserves the Consumer's current whole hypothesis. Original selection retains write authority and incompatible revisions remain necessary.

Complete frozen inputs supply the eligibility witness. The core runtime checks recorded continuation paths; a native correction pipeline confirms inputs of already executing full calls. A call can have a valid input even when its own predicted next state is wrong. The Consumer independently checks its visible whole view. Selected facts, provisional views, and physical execution retain distinct identities through message rejection, durable selection, and recovery.

This paper makes three contributions:

- A feedback-state contract with separate executable witnesses for computation eligibility and whole-view retention across selected roots. Complete inputs, output dependencies, and execution identities determine what survives; a version change alone does not.
- A coordinator that preserves the original selected transition relation while retaining compatible computation and views. Its conditional guarantees distinguish provisional delivery, durable selection, and physical ownership, including supported process failures.
- Actual service evidence with the selected task fixed: factorial controls expose computation-only withdrawal, native correction tests input-confirmed full calls, and fault and sustained-state trials test the coordinator's boundaries and costs.

The headline joint design adds 26.645 s of dual-deadline credit over 714.110 s offered without changing certified histories. A separate 48-run study retains the negative intermediate-load result and distinguishes subsequent availability from first delivery. A matched native comparison holds inputs, models and conditions fixed across the earlier and final implementations. The final protocol raises credit from 30.0 to 65.822 of 66.340 s offered and whole-run passes from zero to six of eight, retaining its two EOS misses.

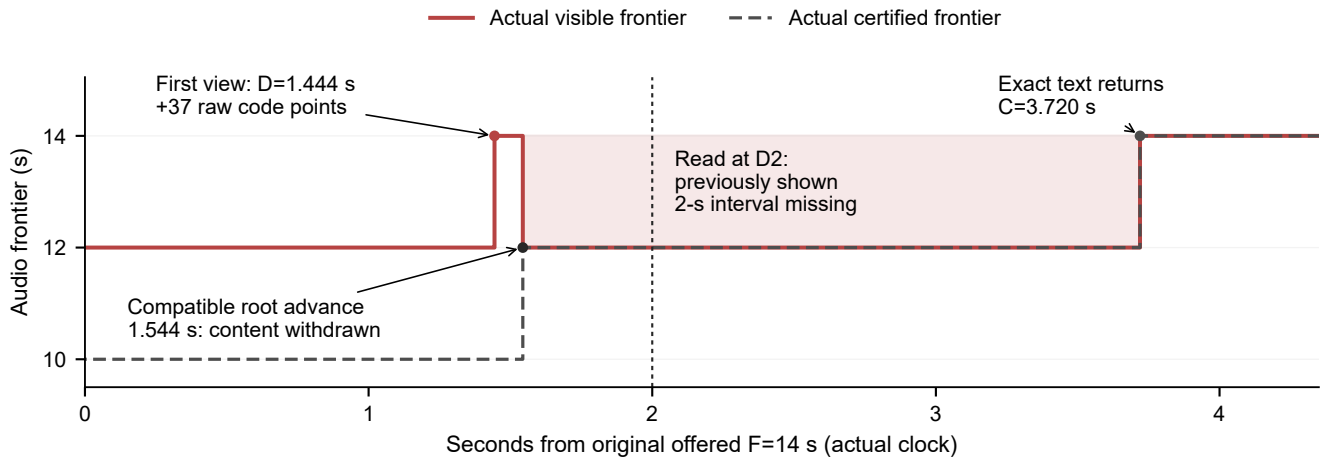



Fig. 1. An actual computation-only interval meets first-display D2 and certified-delivery C8, yet loses its phrase for 2.176 s under a compatible update. The dotted line samples the original D2 read point. This case comes from c16, whose whole-run D2 P99 fails; it is neither a paired counterfactual nor a natural-prevalence estimate.

## II. Feedback Contract

### A. Feedback creates an execution dependency

An immutable audio stream supplies a protected selection path producing *Canonical* facts and a faster *Private* path computing and displaying provisional hypotheses. Selection can change the next call's input: StreamCorrect feeds its chosen correction into subsequent ASR [7]. We coordinate this dependency as the paths advance asynchronously.

For a fixed session and execution contract, write an installed fact as

$$C = (v, F, Z_C, P_C, T_C). \tag{1}$$

Here $v$ is the version, $F$ the certified processing frontier, $Z_C$ the complete selected output, $P_C$ the feedback state, and $T_C$ its bound text. The contract fixes the model, adapter, decoding, audio identity, and transformations to $P$ and $T$. Consumer coverage follows the adapter's publication policy; advancing $F$ need not add characters. Certification means acceptance by protected selection and installation, separately from human-transcription accuracy.

The adapter declares a conservative representation of *all mutable cross-call state that can affect later model inputs, selection, or feedback*. Telemetry and resources are separate. Qwen passes prompt tokens without cross-call numerical KV caches. The native adapter additionally carries audio buffers, token groups, context, alignment position, and decoder configuration. Comparisons require the same contract: text equality cannot replace an omitted cursor or numerical cache. The representation may conservatively include fields unnecessary to a particular call.

### B. Three lifetimes of provisional work

A complete provisional node $n$ records a frozen call

$$q_n = (a_n, P_n^-, b_n, \chi_n), \quad n = (q_n, Z_n, P_n^+, T_n), \tag{2}$$

where $a_n$ is the input-state frontier, $b_n > a_n$ the audio target, and $\chi_n$ binds remaining inputs and execution identity. The model may read the prefix through $b_n$; $a_n$ does not assert token-to-audio alignment. The adapter derives $P_n^+$ and $T_n$ from a complete result. An in-flight call has only its frozen input; partial or cancelled tokens supply no node.

The *dependency lifetime* determines usability under the current root; the *visible lifetime* determines whole-hypothesis retention; the *physical lifetime* ends only when execution finishes or drains. A new root can change the first two before the third ends. Their decisions therefore require separate evidence.

The service retains every accepted input obligation until certified processing covers it. Discarding a branch, requesting cancellation, or recovering an old view does not remove that obligation or reset its arrival clock. In exact mode, a missed deadline never authorizes an uncertified Canonical transition.

### C. Trust and failure boundary

We assume one trusted Owner per session, protected selection, immutable audio and model configuration, unambiguous encoding, collision-resistant digests, and storage honoring successful synchronization. Used calls must faithfully consume their frozen inputs. Host receipts bind API objects, not GPU arithmetic; independent Private and Authority floating-point executions need not agree.

The interface checks identity, completeness, derived state, and ancestry. Native resilience supports worker or Owner-group fail-stop on an intact filesystem; Supervisor, accepted-input ledger, actual Consumer, power observer, and synchronized storage survive. Their original input clocks continue through restart. Supervisor/Consumer failure, host power loss, compromised Owner or Authority, Byzantine code, persistent corruption, and lost storage are excluded. Controlled wrong-input injections in Section V intentionally violate normal Private input fidelity;

we use them to study containment and executor release, not as examples covered by the lawful-continuation proposition.

## III. Valid Continuations and Whole Views

### A. A continuation under a new root

The central question is whether an old call still belongs to a valid provisional continuation after trusted feedback advances. Its birth version records dispatch history. The dependency witness establishes whether it remains usable now.

Let $x = (F, P_C)$. A complete recorded node provides an edge

$$(F, P_C) \xrightarrow{n} (b_n, P_n^+) \quad \text{if} \quad a_n = F \ \wedge \ P_n^- = P_C, \tag{3}$$

under the input, identity, and execution contract of Section II. A finite sequence of such edges is a *valid provisional continuation* of $C$ when each successor consumes the preceding complete output state at its actual frontier. Write $C \rightsquigarrow n$ for a complete path to $n$. A frozen in-flight successor can belong to the same continuation, but contributes no usable result until completion.

**Proposition 1** (Sufficient re-rooting witness)**.** *Suppose an old branch contains a complete boundary node at the new certified frontier $F$, with output state $P_C$. If its subsequent recorded calls form an unbroken input–output path to the actual branch tip, those nodes are a valid provisional continuation of $C$. An in-flight successor may continue when its unchanged frozen input is that tip and its target exceeds $F$.*

*Proof.* The boundary equals $(F, P_C)$, so the first successor satisfies Eq. (3). Each subsequent call consumes the preceding complete output at its recorded frontier under the same contract. Induction reaches the tip and the unchanged input of its in-flight successor, which supplies no output yet. □

This establishes dependency validity, not Canonical write authority. Provisional results remain revisable. The witness follows actual calls without assuming bitwise equality of different floating-point executions.

A branch born under $v$ can reach $(F, P_C)$ when $v+1$ installs, making version-only rejection wasteful. Conversely, equal text can hide different tokens or alignment state; equal state at a different frontier also fails Eq. (3).

### B. Retaining computation: $R$

The core Qwen guard conservatively implements Proposition 1. On installation, it requires a complete Private node at exactly $F$ with full feedback tokens $P_C$, then checks frozen inputs and frontiers along complete successors to the actual tip. An intermediate match is insufficient. An in-flight successor must be the unique tracked task, consume that tip, and target audio beyond $F$.

Success retains the branch without changing job IDs, birth versions, inputs, or results; otherwise the Owner fences it and rebuilds from an allowed state. Boundary-node and unique-slot requirements restrict this sufficient guard, not every valid reuse scheme.

### C. Retaining an actual view: $S$

A snapshot Consumer can replace a valid ahead view with a shorter certified snapshot despite computation retention. RevTrace maintains separate certified and provisional whole-hypothesis lanes, without an uncertified suffix splice or certified-prefix status for retained views.

For a Canonical application, $S$ checks the complete visible node's input directly against $(F, P_C)$ or follows complete published ancestors to that boundary. Missing, cyclic, or ambiguous ancestry is rejected. Unlike $R$'s conservative guard, a directly rooted published call needs no extra Private boundary node.

Before actual application and its receipt, the kernel checks the preview against the durable root, registered results, dependency path, and bound text/frontier. A caller-supplied compatibility flag cannot replace these records.

**Proposition 2** (Continuity at a compatible application)**.** *Immediately before an authorized Canonical application, let the visible whole provisional view $h$ be ahead of its new frontier $F$. If $S$ proves $C \rightsquigarrow h$ and no independent withdrawal mode is invoked, that application advances the certified lane and preserves $h$ unchanged.*

*Proof.* The authorized compatible-and-ahead branch installs $C$ in the certified lane while retaining $h$'s node, text, and frontier. That application therefore withdraws none of $h$. □

Incompatible updates must withdraw the view, and later valid results can revise it. This is a per-application property, not global monotonicity. Snapshot consumption and conservative recomputation can be safe; the four-arm study measures the extra progress and cost of joint retention.

### D. Native feedback and conditional full calls

StreamCorrect's original recognizer, learned correction, and feedback update provide a concrete instance of the contract in Section II [7]. Let $P$ be a complete closed native state and $X$ the next accepted audio input. We expose the recognition boundary as

$$A(P, X) = (B, M), \tag{4}$$

where $B$ contains the ordered beam candidates and $M$ contains the live engine state and inputs required by the remaining correction and update. The protected original $C/U$ selects a correction and produces the next closed state. Its selection, candidate order, prompts, returned token IDs, and update rules remain intact.

Before that selection finishes, the original update tail supplied with a typed first-beam identity prediction produces a provisional closed state $\widehat{P}^+$. This projection has explicit provenance: recognition has executed, but learned correction has not completed for this projected node. It supplies a conditional input for the next complete call. Both GPUs host the original full recognizer–corrector pipeline, so a call dispatched on this input executes the protected original $C/U$ as well. Completed conditional full results and typed identity projections

remain distinct objects. Neither is installed merely because its predicted text looks plausible.

When the selected parent arrives, the Owner compares the call's entire frozen start state, actual frontier, execution contract, and session epoch with the installed parent. An exact match confirms the already owned call: an in-flight call continues, or an already completed result becomes eligible for ordered installation. The job's birth root, identifier, epoch, inputs, and execution history are unchanged. The call's own identity prediction need not equal its eventual corrected output; confirming its input establishes eligibility, while original correction determines its result.

An earlier ancestor mismatch removes support for the tracked provisional chain, but does not prove that a later complete input cannot converge. Cancellation before correction therefore requires an actual selected state at the call's same start frontier that disagrees with its complete input, coverage of its target without remaining use, or a changed epoch. A still unresolved future parent supplies none of these proofs. Loss of current view support can therefore require withdrawal without yet authorizing cancellation of that unresolved call.

This direct native witness instantiates the continuation relation of Section III. It differs from the core Qwen computation guard, which conservatively requires an explicit matching boundary node and a uniquely tracked successor. These are different sufficient implementations of eligibility. The whole-view Consumer likewise keeps certified and provisional provenance separate and applies complete supported hypotheses. The native service comparison uses two full-pipeline workers in both arms and differs in conditional dispatch and retention. Ready-only ASR handoff is a separate developmental control. These are author-built implementations of the feedback principles; the measured benefit concerns confirmed full calls.

### E. Preserving the protected program's semantics

Let $P \Rightarrow_{\mathcal{A},X} P'$ denote a transition allowed by the original protected program on accepted immutable input $X$. The relation may be nondeterministic. A complete original call includes its original selection and update, or its declared buffer-only path; a typed identity projection is not such a result. We project a protocol execution onto its distinct durable selected facts, deleting speculation, waiting, local rejection, cancellation, collection, and repeated delivery. This restricted projection follows the standard stuttering view of refinement [8]; provisional whole-view behavior remains outside its alphabet and has a separate compatibility property.

**Proposition 3** (Selected-trace refinement)**.** *Assume sufficient declared state, faithful execution of frozen inputs, exclusive serial selection, correctly bound message identities, and storage that honors successful synchronization. If each new durable fact binds a complete original call whose full input equals the preceding selected cut, retains the call's original identity and valid admission authority, and is synchronized before installation or certified application, then the projected facts form an execution allowed by $\mathcal{A}$. Replaying an existing fact adds no transition.*

*Proof.* The initial cut is an initial state of $\mathcal{A}$. Deleted protocol actions do not extend the selected prefix. At each new fact, the guard identifies a complete original call on exactly the preceding cut and accepted input. Its actual result is therefore an allowed next edge, irrespective of when the call began. Synchronization records that edge before either privileged application. Recovery verifies the same chained prefix and replays its existing identities. A fact synchronized just before an Owner crash can first become visible during replay, but is still counted once. Induction proves the claim. □

The durable append is the selection boundary; a later installation or Consumer application does not select again. An accepted complete buffer can retain its producer identity after that worker exits. This differs from admitting a new late message after its authority has been fenced. The certified Consumer frontier derives only from the checked fact prefix; an ahead whole view remains provisional. Thus input confirmation permits early execution without requiring every predicted next state to be right.

A separately frozen specification explores 6,231 two-frontier normal states and 14,074 one-frontier failure states, each to a 12-event depth, with 27 named controls. An independent trace checker binds actual facts and whole-view applications to original call receipts. The specification is not extracted from the runtime; depth bounds truncate exploration, and neither check verifies all Python, CUDA, or operating-system behavior. Exact equality with a fixed neural reference is separate empirical evidence, rather than a premise that arbitrary GPU reruns must be bitwise identical.

## IV. Execution Protocol

### A. Complete results and physical ownership

In the core runtime of Figure 2, dispatch binds session, incarnation, job, input, target, and contract. A row returns either a complete result with re-derived state/text or cancellation without partial state. Complete rows can be installed and consumed before their batch finishes. Only all terminal rows and device synchronization authorize `BatchDrained` and executor reuse. The native process lifecycle below has separate evidence.

### B. Covered work and safe-point exit: $K$

For the tested use model, $K$ requests exit when the installed Canonical frontier covers a proposal's target and it has no remaining declared use. This differs from $R$: valid work beyond the new root can remain useful. Workers observe `CancelRequest` at decode boundaries; completion can win, but each row has one terminal outcome. Static batches are not compacted or refilled; masked costs and drain delays remain measured.

### C. Durable facts and outstanding input

Before acknowledgment, synchronized frames bind parent/next states, request identity, selected complete result, and receipt; recorded deltas must reconstruct the state. Group-ready commit shares synchronization among validated sessions

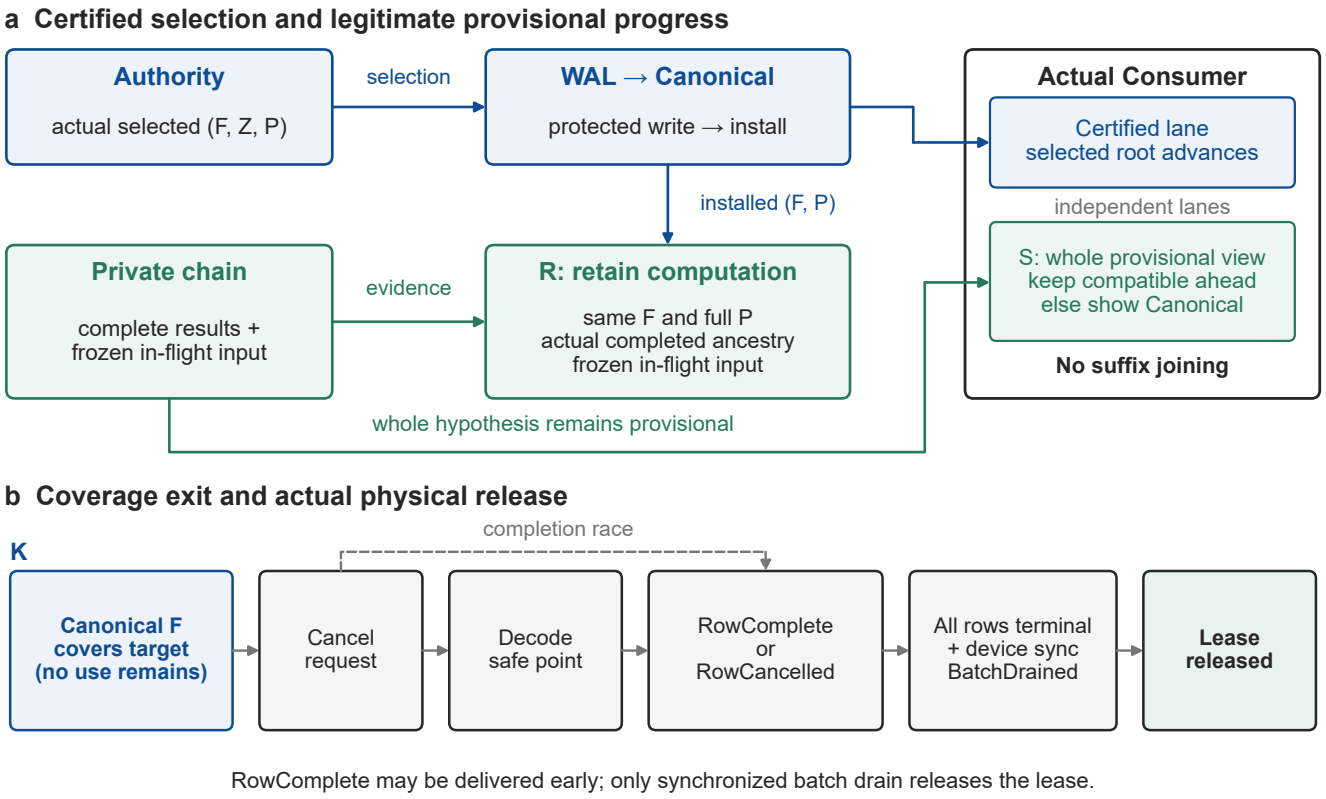



Fig. 2. The core RevTrace runtime coordinates complete provisional computation and whole Consumer views with certified state. $R$ and $S$ use the same declared feedback dependencies, with separate sufficient guards. Coverage retirement uses a different predicate. A complete result may be consumed before its batch drains; only terminal rows plus device synchronization release the physical lease.

without a collection timeout or cross-session atomic transaction. Uncertain append/application outcomes block writes. Write-ahead recovery [9] reconstructs facts and identities; the native extension binds this to original correction calls.

### *D. Validation boundaries*

Propositions 1 and 2 concern the stated guards and compatible applications. The artifact retains an earlier, separately frozen recovery prototype and its bounded model and crash results; they do not validate the later core or native fault paths. The selected-trace argument of Proposition 3, its finite specification, and actual execution checks have distinct scopes.

### *E. Continuing through delivery and process failures*

Each job has an Owner namespace and monotonic issue number; each physical slot has a worker incarnation, following established RPC identity and duplicate-suppression principles [10]. Fencing an incarnation removes message authority without renaming its already accepted complete results.

*a) Local rejection and physical release:* The ingress guard classifies events before changing busy slots or selected state. Exact duplicates and fenced messages are local rejections that cannot release newer leases. Contradictory currently owned contents freeze writes; unknown exceptions remain failures. Per-incarnation Pipe endpoints and receiver threads isolate incomplete logical frames from the Owner loop. Replacement requires process exit, CUDA release, joined receivers, and strict validation of complete receipts on the sealed channel. The Owner then fences the incarnation and retires unfinished work. Accepted complete buffers and supported whole views survive producer death with their original identities.

*b) Durable selection and recovery:* Before installation, a synchronous hook records complete parent/next states, original correction and IDs, job identity, Owner/session epochs, and producer incarnation. Framing, checksums, and chaining bind the fact; synchronization precedes installation and publication. Uncertain writes freeze the writer. Recovery may discard only an incomplete final frame; complete corruption fails. These conventional write-ahead rules [9] bind the original feedback transition.

The surviving Supervisor continues the input ledger and original clocks under the failure boundary of Section II. After confirming old physical ownership has ended, it advances the Owner namespace, replays the checked fact prefix, and cold-starts workers. Replay adds no inference; old publications gain no new authority. Surviving provisional views remain subject to subsequent root checks. Both schedulers share this coordinator, durability, recovery, and actual Consumer; only conditional

full-call dispatch and retention differ.

*c) Conditional continued service:* For finite accepted work after a finite supported failure prefix, progress requires readable durable storage, eventual physical release and loading, terminating original calls that advance pending frontiers, fair Owner/Consumer scheduling, and unexpired recovery/service budgets. Under these conditions, calls from the actual cut can discharge remaining obligations without accepting fenced messages. This gives no RTO or D2/C8 bound or unbounded liveness proof: cold loading, recovery, and recomputation remain measured against original clocks.

*d) Control history:* Current roots and live-call, complete-buffer, visible-view, and outbox dependencies remain pinned. Actual Consumer acknowledgment releases publication references, while the visible view stays pinned. Namespace and issue high watermarks prevent ID reuse after collection. Unreferenced terminal fingerprints and audit windows have fixed limits; full diagnostics stream to disk. These are protocol-entry bounds relative to active dependencies, not constant total memory: immutable input, current state, cumulative text, and offline analysis remain separate.

## V. Evaluation

### A. Experimental design and service endpoints

We evaluate two service obligations under unchanged selected feedback histories: retaining dependency-valid progress already at the Consumer, and completing selected transitions before their original deadlines. The core factorial study isolates computation and whole-view retention; the native study tests input-confirmed full execution of the original correction pipeline. Timing, content support, and energy remain separate. The core experiments use actual neural calls, durable input and Canonical journals, and an instrumented FIFO Consumer. Measurements use Consumer applications rather than publication or display replay. Accepted input obligations retain their original arrival clocks.

*a) Implementation and workloads:* The service uses two 24-GB RTX 3090s: an FP16 Private worker and an FP32 Authority worker. Qwen3-ASR 1.7B [11] is the primary backend; Whisper base.en [12] tests a second adapter. The common carrier uses complete-row delivery, at most 16 rows per batch, a 256-token generation limit, the protocol kernel, and synchronized journals. These experiments use `row_complete`: physical early exit is disabled, so the effects of $R$ and $S$ cannot be attributed to cancellation. Model startup is outside the serving interval and is recorded separately. Serving energy integrates both physical boards; CPU energy is outside that measurement.

The independent confirmation comprises 24 previously unused speaker streams from LibriSpeech [13], totaling 714.11 s, selected using metadata before inference and partitioned into three eight-speaker cohorts. Its 110 cells contain 60 primary cells (three cohorts, two backends, two arrival patterns, five arms), 20 fresh-process repeat cells on cohort A, and 30 coarser-feedback controls. Primary Authority advances at every 2-s input boundary ($\rho = 1$); $\rho = 3$ changes both selection frequency and feedback segmentation. Synchronized streams start together; staggered streams start 0.25 s apart. Repeats and arms reuse these sources: the independent acoustic sample is 24 speakers, not 110 deployments.

*b) Comparators:* The four arms cross computation retention $R$ with whole-view retention $S$ (Section III). With $R$ disabled, computation rebuilds from the selected root rather than retaining its witnessed continuation. With $S$ disabled, a Canonical application replaces the provisional view with the new certified snapshot. This *checked-snapshot* Consumer still validates Private dependencies against the same root and passes the same protocol kernel. Removing retention does not remove validation, durability, input accounting, or the execution capability of Section IV. The fifth arm, *Full composition (ours)*, is a complete-composition positive control: an alternative policy instantiation with the same retention relation, Consumer protection, and shared execution/commit substrate. It is neither an external pre-existing implementation of our complete design nor an independently reimplemented neural stack. Comparable behavior is the expected positive-control outcome.

*c) Endpoints:* For offered interval $i$, let $w_i$ be its audio duration, $r_i$ its original ready time, $d_i$ its first actual visible covering time, and $c_i$ its first actual certified covering time. Original joint-deadline credit is

$$A_{2,8} = \sum_i w_i \, \mathbf{1}\{d_i - r_i \le 2 \ \wedge \ c_i - r_i \le 8\}. \qquad (5)$$

All offered audio, including end-of-stream (EOS) tails and misses, stays in its denominator. Whole-run service passes only when display P99 is at most 2 s and certified-view P99 at most 8 s. P99 is computed over offered frontiers within each run; credit weights their audio durations. Thus a P99 pass can coexist with a small tail miss. We separately measure adjacent visible-frontier withdrawals, their audio magnitude, and the integral of the deficit below the previously observed high watermark. Text revisions, final word error rate (WER), and board energy remain separate endpoints. Independent recounters check complete selected $F/Z/P/\mathrm{text}$ histories and reconstruct dependency witnesses from actual complete outputs and consumed inputs, rather than trusting a retention flag.

### B. Independent confirmation of the composition

Table I reports the pre-specified Qwen, $\rho = 1$, staggered stratum, including all three cohorts. Retaining computation alone raises deadline credit from 654.115 to 664.150 s but increases audio-frontier withdrawals from 12 to 17. The additional work survives the new root, while snapshot consumption still discards its visible progress. Retaining views alone reduces withdrawals to one but yields 628.435 s of credit. The joint arm obtains 680.760 s and zero audio-frontier withdrawals: 26.645 s, or 3.73 percentage points of offered audio, above $R_0 S_0$. Its gain over computation-only is 16.610 s.

The cohort-wise credit interaction, $I = Y_{11} - Y_{10} - Y_{01} + Y_{00}$, is $+18.530$, $+6.205$, and $+17.555$ s. These three contrasts share queues within each cohort. They support a composition effect in this stratum, rather than treating individual deadline

TABLE I
INDEPENDENT QWEN CONFIRMATION, $\rho = 1$, STAGGERED ARRIVALS. EACH ARM OFFERS 714.11 S ACROSS THREE EIGHT-SPEAKER COHORTS. WITHDRAWALS AND CREDIT ARE TOTALS; P99 AND ENERGY ARE MEDIANS OF THREE PER-RUN STATISTICS. ALL 15 RUNS MISS THE WHOLE-RUN D2 CRITERION.

| Arm | Audio withdrawals | Display P99 (s) | D2/C8 credit (s) | Energy (J/audio-s) |
|---|---|---|---|---|
| $R_0S_0$ | 12 | 3.797 | 654.115 | 59.72 |
| $R_0S_1$ | 1 | 3.891 | 628.435 | 59.16 |
| $R_1S_0$ | 17 | 2.431 | 664.150 | 59.68 |
| $R_1S_1$ | 0 | 2.329 | 680.760 | 59.37 |
| Full (ours) | 0 | 2.534 | 667.530 | 59.73 |

windows as independent trials. The joint arm still erases 569 visible characters: zero audio-frontier withdrawal does not mean immutable text. Its 2.329-s median display P99 also remains above D2. The result preserves more timely progress without establishing that this workload meets its whole-run service target.

The full composition likewise has zero frontier withdrawals, the same 569 erased characters, and comparable timing and energy. Its lower credit than the joint arm in this stratum is not evidence that a different complete composition is inherently inferior. Across the confirmation, all 144 matched source/backend/granularity/arrival history groups agree on complete Canonical $F/Z/P/\mathrm{text}$; all 22 paired full-composition comparisons preserve those histories. The ablations therefore isolate how legal progress is retained, rather than buying latency by changing the selected feedback task.

### C. Relative progress explains when retention matters

The useful opportunity is an already observed whole view ahead of a new Canonical frontier whose complete dependency path remains valid. In the headline joint arm, 18 of 368 Canonical updates encounter such a view and retain it. Computation-only encounters 17 and clears all 17. Under synchronized arrivals the joint arm encounters only three of 368; its credit is 0.790 s below computation-only. Without an ahead view, preserving it has little service value. These are witnessed incidences in the executed workload, not a production prevalence estimate.

Timing can also change the ranking. In the fresh-process staggered repeat on cohort A, joint credit is 226.320 s versus 230.565 s for computation-only, and $I = -2.245$ s. We retain this repeat alongside the three positive primary contrasts. The combination's value depends on relative progress and actual feedback compatibility; independent source confirmation does not imply an invariant timing advantage.

### D. Operating workloads and subsequent timely continuity

We next freeze three development workloads—eight synchronized sessions, eight staggered sessions, and sixteen staggered sessions—with four arms and three fresh processes per arm (36 cells). A single later exploratory midpoint adds twelve staggered sessions and the same twelve arm/repeat cells. Figure 3 retains all 48 runs. The nested source lists offer 121.205, 185.300, and 250.180 s per eight-, twelve-, and sixteen-session run. Adding sessions also adds source content; these are explicit operating workloads, not a pure concurrency capacity curve.

All arms pass whole-run D2/C8 in the two eight-session workloads. Joint display P99 is 1.426–1.528 s when synchronized, where the arms largely converge. Staggered joint P99 reaches 1.955 s, and an earlier development run reached 2.034 s, exposing a narrow timing margin. Two 0.215-s tails exceed D2 in the new eight-session runs and remain uncredited even though their runs' P99 passes. At sixteen sessions, all twelve runs fail D2, while C8 passes. Joint credit exceeds $R_0S_0$ by 20.280/22.885/20.180 s, but its credit interactions are $+20.910$, $+26.380$, and $-9.575$ s. Against computation-only, joint credit changes by $+18.910$, $+5.055$, and $-7.225$ s, and display P99 improves in only one repeat. Continuity is the stable distinction: computation-only withdraws compatible views 16/17/12 times, versus zero for joint.

First credit does not reveal a later withdrawal. A frozen secondary recount therefore follows intervals already delivered within D2 whose eventual certified covering view also meets C8. At each subsequent Canonical application it classifies withdrawal using the complete old view's dependency witness, separately recording necessary incompatible withdrawals. Loss lasts until an actual covering view returns or the next Canonical boundary censors this root's window. At sixteen sessions, computation-only still has 10/12/9 timely compatible loss windows with 24.903/23.635/15.736 audio-s×wall-s deficit area; joint has none. Such holes intersect the original D2 read point for 6.000/10.675/4.000 s of distinct audio. The secondary measures continuity after delivery; it neither changes Eq. (5) nor certifies provisional content.

The twelve-session witness is *not established*: all twelve runs fail D2, and computation-only never loses a timely complete non-EOS interval satisfying the frozen witness criterion. Joint has no compatible loss windows. The predeclared stop rule ends this search; full criteria and per-run results remain in the artifact.

### E. Quality and cost boundaries

All headline Qwen arms have WER $26/1908 = 1.36\%$ and identical selected histories. Retrospective support counting first-certified content plus first-ahead whole views that exactly match later same-$F$ Canonical text leaves the headline credits unchanged. Elsewhere raw differences remain: joint loses 2 s of strict support in the sixteen-session third repeat and 2 s in each of the midpoint's first two repeats, although the frozen text normalizer agrees. This comparison is with the selected model output, not human-transcription correctness. Whisper exposes a larger adapter boundary: WER is 83.12% at $\rho = 1$ and 11.53% at $\rho = 3$, whose different feedback contract prevents interpreting it as a quality-preserving speedup.

The headline arms consume similar board energy (Table I); energy does not rank them in credit order. At sixteen development sessions, joint reduces total model time by 0.66–1.59 s against $R_0S_0$, while serving wall time does not consistently

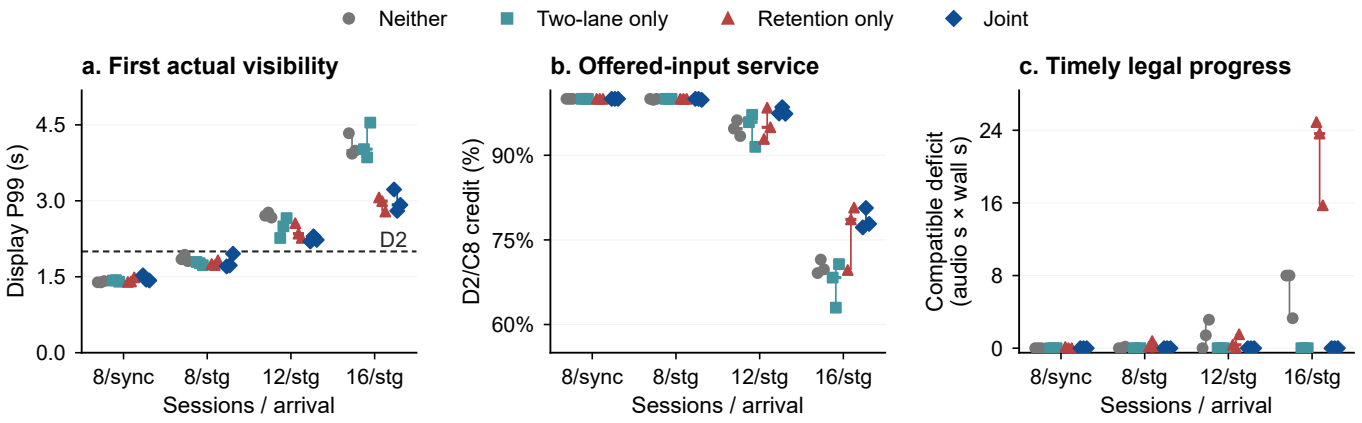



Fig. 3. All four development workloads and three fresh-process runs per arm, including the negative twelve-session midpoint. Points are real per-run values; ticks and ranges describe their median and observed range. Panel (c) is the separate secondary measure of subsequent compatible loss of originally timely intervals, censored at the next Canonical root when necessary. It does not replace panel (b)'s original first-observation credit.

fall and the three energy differences change sign. The measured benefit is the progress delivered and preserved under a fixed feedback contract. It is not a general reduction in inference cost or a cure for overload.

### *F. Native correction with the original selected task*

We retain StreamCorrect's published recognizer, learned correction, and feedback update [7], using the fixed 500-ms, no-VAD adaptation and original FP32 execution. Both compared policies have two protected full workers, the same ASR early-view Consumer, and identical neural warmup. Conservative execution waits for selected parent feedback before starting the next call; full-call retention confirms conditional calls as in Section III-D. We distinguish the earlier frozen native implementation (`bdfa3b13`) from the resilience implementation (`4f4c3cab`). Historical independent confirmation and the new matched comparison retain their own identities.

*a) Development and frozen confirmation:* Development on AISHELL-1 [14] tests original full and buffer-only steps, including projected ends that differ from selection without invalidating their inputs. Full calls, unlike ready-only ASR handoff, retain still-executing work. The artifact retains the pilot and development matrix, including two 0.068-s EOS misses.

For the earlier `bdfa` independent confirmation, the method was frozen before metadata selection of four unused AISHELL-1 speakers in an 8–9-s band; all initial selections remain. Two policies and two fresh-process repeats give 16 runs, with randomized pair blocks, reversed arm order, and swapped GPU roles. Speakers are the independent acoustic units. Every original 500-ms frontier and partial EOS stays in the denominator.

*b) Task fidelity and timely service:* Offline original full calls supply 68 reference transitions, separately from service timing. All 272 Canonical transitions in this confirmation reproduce their complete original start and end states, $C$, and returned IDs, including the implementation tag. Both arms have five character edits over 81 unique reference characters, or 6.17% CER. This measures the adaptation, not the published author's benchmark CER.

TABLE II
MATCHED VERSION COMPARISON: FOUR PREVIOUSLY MEASURED SPEAKERS, TWO POLICIES, TWO VERSIONS, AND TWO FRESH TRIALS (32 RUNS). EACH ROW OFFERS 66.340 S. CREDIT AND BOARD ENERGY ARE TOTALS; D99 IS THE MEDIAN OF PER-RUN VALUES. ENERGY USES ONE COMMON EXTERNAL ORIGIN-TO-CONFIRMED-PHYSICAL-MODEL-DRAIN WINDOW, AN UPPER OBSERVATION BOUND. THIS IS A MATCHED REGRESSION STUDY, NOT NEW ACOUSTIC CONFIRMATION.

| Version | Policy | Credit (s) | D99 (s) | Energy (kJ) | Pass |
|---|---|---|---|---|---|
| `bdfa` | Conserv. | 33.500 | 4.450 | 44.690 | 0/8 |
| `bdfa` | Full-call | 66.340 | 1.633 | 39.735 | 8/8 |
| `4f` | Conserv. | 30.000 | 4.759 | 47.019 | 0/8 |
| `4f` | Full-call | 65.822 | 1.638 | 43.952 | 6/8 |

The earlier independent confirmation on `bdfa` raises credit from 37.0 to all 66.33975 s and whole-run passes from 0/8 to 8/8. We then freeze a matched 32-run comparison on those same four sources, with common FP32 models, neural warmup, allocator, swapped GPU roles, and external board-power observation. All 544 selected transitions reproduce complete original inputs/outputs, $C$, and IDs under the declared static code-version mapping. Table II reports these new runs. The final implementation raises credit from 30.0 to 65.821625 s and passes from 0/8 to 6/8. Its two failures are S0908 EOS intervals: paired display P99 increases by 0.163/0.050 s against `bdfa`. Other paired differences have both signs; similar medians do not establish equivalence. Through confirmed physical model drain, final Full-call uses 6.52% less board energy than final Conservative and 10.61% more than earlier Full-call. Model drain and source-specific service cleanup have different endpoints; separate EOS-window and original per-version results remain in the artifact.

*c) Resident independent requests:* A new finite contract keeps the models and workers resident while a single client submits four pre-registered, independent requests. Each original file EOS closes its request; the next origin is assigned only after actual certified EOS consumption. This is closed-loop request service, not online VAD, one continuous recording, or open-arrival capacity. Two policies and two fresh trials preserve all 272 original transitions and per-request input clocks. Both worker PIDs persist across each trial's requests, with one initial model load. Full-call meets D2/C8 on 7/8 requests versus 0/8, earning 64.118875 versus 31.0 of 66.33975 s offered. The second repeat's S0908 request has D99 2.229 s; residency does not establish an invariant deadline margin.

*d) Retained work and visible content:* The earlier independent confirmation contains 92 promoted calls whose recognition begins before their selected parent finishes; 20 have their own predicted end differ from original correction. Twenty-four invalid or covered calls stop after ASR, and two completed wrong-input results are discarded, without changing selection. This separates input confirmation from requiring correct future predictions.

One original-source Canonical update from 4.0 to 4.5 s keeps a supported 5.0-s view and its executing call. Two necessary updates withdraw 6.0 to 5.5 s when the actual parent refutes the complete input. These witnessed cases do not estimate another $R \times S$ interaction or natural rate.

In that earlier confirmation, first whole-text agreement with later same-$G$ Canonical text supports 29.0 versus 51.518125 s of credited audio; 14.821625 s of Full-call credit lacks support, with zero unknown coverage. Raw and whitespace-only comparisons agree. Unchanged-text rows cover 10.0 versus 17.0 credited seconds. Frontier credit is neither newly transcribed characters nor human correctness.

An earlier two-source capacity comparison remains negative: retention earns 17.0 s versus 18.5 s for independent full replicas, at higher energy and with a different warmup/Consumer. Single-chain gains do not establish superior saturated multi-source capacity.

### G. One coordinator through faults and sustained state

The failure and state study uses `4f4c3cab`, the resilience version also compared in Table II, with separately frozen workloads. Both policies share durability, message guards, recovery, Consumer, original FP32 models, and the `expandable_segments:True` allocator setting. Initial loading and fixed warmup precede the serving clock; replacement loading, replay, recomputation, and final physical drain remain inside it.

The short bank has six conditions, two policies, and two fresh processes on one development clip (24 trials), plus four EOS/compatible-view regressions. Across these and eight sustained trials, an independent checker binds all 3,976 durable selected transitions to original complete call receipts and actual Consumer applications. It finds no illegal installation or incorrect owned-message rejection, and preserves every accepted input and its original clock. Eight injected duplicate/late messages are locally rejected without Canonical mutation; eight worker-only deaths have confirmed physical replacements. Short reference comparisons map only the static code-version tag; long comparisons are unprojected. Conditional calls contribute 354 durable facts; 352 have an original Owner installation record, and two first reach the Consumer through recovery. There is no completed-buffer installation in these trials. In four worker-after-ASR cases, an already consumed whole view remains in Consumer state across its producer's exit and fencing until legitimate Canonical correction. This is retained availability, not another delivery of the old job or a claim that its prediction was correct.

Table III retains deadline failures. Normal Full-call display P99 is 1.798–2.053 s: one of two fresh trials passes, while the other misses D2 only at EOS. Duplicate/late delivery has the same one-of-two boundary. All worker-fault trials fail D2 but pass C8; all Owner-fault trials fail both. After durable selection, old facts reappear in 1.36–1.81 s, whereas new certified post-fault completions take 64.61–108.08 s across the Owner cases. A surviving worker can provide a post-fault completion before replacement readiness. Neither replaying a fact nor finishing the task establishes fault-time D2/C8.

*a) Complete results across producer death:* The 36 trials do not install a completed conditional buffer. We therefore freeze two matching-input and two mismatching-input timing cases on a development source. A transport fixture holds the parent after original ASR and the child after sending its complete original result. After the Owner registers that result, we kill its producer, check actual exit, CUDA release, sealed-channel validation and incarnation fencing, then release the parent's original correction. All four executions complete their protected task. Independent reconstruction confirms that both matching cases install and consume the original result once, preserving its job, birth and incarnation; both mismatching cases reject it using the actual same-frontier parent input. The positive cases also complete a subsequent independent request. These are controlled interleavings, not natural incidence or fault-time SLO evidence. A separately frozen, zero-NN reconstruction validates these unchanged runs; their original audit gates remain failed and are retained in the artifact with the READY indexing defect.

*b) Controlled sustained-state stress:* Two metadata-selected AISHELL-1 speakers supply 24 utterances each, concatenated as exact PCM into 228.51025-s and 208.1003125-s synthetic streams without resets or EOS at joins. A continuous original-engine reference without between-step restoration and 875 separately restored steps agree on complete start/end state, $C$, and IDs, crossing 398/357 actual buffer shifts and 353/252 context changes. Two policies and two fresh processes yield eight trials and 3,500 matching selected steps. Unpinned terminal receipts and each retained audit class stay at or below 64 entries; live dependencies, serialized state/output, Owner RSS, and cumulative disk logs remain separate.

This is state stress, not viable continuous ASR: all eight trials fail D2/C8, with display P99 of 449.727–502.952 s. Both policies reproduce the original task's severe insertion

TABLE III

SAME-SOURCE SHORT FAULT STUDY: SIX CONDITIONS ON THE FIXED DEVELOPMENT CLIP, TWO FRESH TRIALS PER POLICY. EACH CELL LISTS CONSERVATIVE ABOVE FULL-CALL; RANGES ARE OBSERVED MINIMA AND MAXIMA, NOT CONFIDENCE INTERVALS. ILLEGAL; DONE GIVES ILLEGAL INSTALLATIONS AND COMPLETED/2 TRIALS. REJECT/MUTATION COUNTS LOCAL REJECTIONS AND RESULTING CANONICAL CHANGES. NEW RESULT MEASURES FAULT TO ACTUAL CERTIFIED CONSUMPTION OF A NEW DURABLE FACT WHOSE ORIGINAL CALL COMPLETES AFTER THE FAULT; IT EXCLUDES OLD-FACT REPLAY AND REPLACEMENT READINESS. D2/C8 COUNTS JOINT WHOLE-RUN PASSES; EOS COUNTS D2 MISSES. BOTH-BOARD ENERGY SPANS THE ORIGINAL SERVING CLOCK THROUGH PHYSICAL DRAIN.

| Condition | Illegal; done | Reject / mutation | New result (s) | D2/C8 pass | EOS miss | Energy (J) |
|---|---|---|---|---|---|---|
| | 0; 2/2 | 0 / 0 | | 0/2 | 2 | 6566–6574 |
| No fault | 0; 2/2 | 0 / 0 | – | 1/2 | 1 | 5896–5911 |
| | 0; 2/2 | 4 / 0 | | 0/2 | 2 | 5701–5893 |
| Duplicate / late | 0; 2/2 | 4 / 0 | – | 1/2 | 1 | 5538–5846 |
| | 0; 2/2 | 0 / 0 | 1.90–2.22 | 0/2 | 2 | 5182–5401 |
| Worker after ASR | 0; 2/2 | 0 / 0 | 2.15–2.21 | 0/2 | 2 | 5447–5647 |
| | 0; 2/2 | 0 / 0 | 2.10–2.25 | 0/2 | 2 | 4775–5178 |
| Worker partial frame | 0; 2/2 | 0 / 0 | 1.76–2.23 | 0/2 | 2 | 5492–5594 |
| | 0; 2/2 | 0 / 0 | 86.42–108.08 | 0/2 | 2 | 24549–28019 |
| Owner before durable | 0; 2/2 | 0 / 0 | 64.61–107.09 | 0/2 | 2 | 19571–26557 |
| | 0; 2/2 | 0 / 0 | 83.90–87.05 | 0/2 | 2 | 22415–24200 |
| Owner after durable | 0; 2/2 | 0 / 0 | 101.80–104.44 | 0/2 | 2 | 26483–27171 |

errors: CER is 3103/486 (638.48%) and 1313/522 (251.53%) on the two streams. Full-call also uses more board energy. The original decoder's length guard can skip its decoding loop, so frontier credit need not add decoded text. The checker observes 90 compatible-ahead applications, all preserved, and 1,118 necessary incompatible withdrawals across the 36 trials. These are controlled coverage counts, not natural prevalence or transcription correctness.

*c) Input confirmation and actual use:* A separate read-only audit pairs all 5,529 actual dispatches with the selected complete cut at their start frontiers. Of 1,901 conditional dispatches, 360 inputs eventually match, 1,541 are refuted, and none remain unresolved. Matches are 40/57 on normal short traces, 58/90 under controlled faults, and 96/924 and 166/830 on the two no-reset sources. These strata have different workloads and acoustic units; their pooled ratio is not a production estimate. Six matched conditional jobs fail to become durable after injected process failures. Thus eventual input equality, live ownership, durable selection, installation, and actual consumption require separate accounting.

Own predicted-end disagreements (409/457 and 333/416 observed predictions) are selected-call diagnostics, not the dispatched-input denominators. The first observed D2/C8 misses and a conservative offline excess-length bound both precede the after-trim 448-token condition on each stream. The length bound allows all reference characters in the currently started utterance, including future ones. It is neither online prefix CER nor a reset signal, and the observations do not identify a unique causal mechanism.

### H. From physical release to supported service

A separate 24-run core study crosses $R$ and $K$ with $S$ enabled, two arrival patterns and three fresh processes, using a frozen wrong-conditioning fixture. Selected histories and WER $14/687$ match; all 675 same-input/prompt/audio/peer-order pairs return identical outputs. Synchronized $R1$ releases the next 16-row computation 473–577 ms earlier. All 96 following rows are actually consumed and their whole texts match later same-frontier Canonical text. Only 6/6/1 rows switch from deadline miss to hit. Adding $K$ changes credit by +7.970 to +15.255 s when synchronized and $-26.000$ to $+21.780$ s when staggered, without consistent energy savings. Request acknowledgment still precedes physical drain. This is a measured release-to-service opportunity, with no stable positive $R \times K$ interaction.

## VI. RELATED WORK

*a) Incremental information and dependency reuse:* The incremental-unit model separates newest committed information from the active right edge and propagates dependency revocation [1]. Its commit promises that a unit will not be revoked. Here, certification records acceptance by the protected selection protocol; later selected facts may still revise the transcript. Adapton checks recorded dependencies; Nominal Adapton combines names with arguments and dependency checks to recover reusable work [2], [3]. Differential Dataflow maintains incremental computations across evolving versions [4]. These works establish much of the basis for dependency-aware reuse. RevTrace instantiates that basis for complete neural feedback state and frozen executed calls. Its service obligation connects execution eligibility to the actual whole view retained when an independently selected root is applied. Our sufficient witness follows recorded executions; it does not transfer a from-scratch theorem to independently repeated floating-point inference.

*b) Revisable speech services and speculation:* Whisper-Streaming converts an offline recognizer into incremental service using LocalAgreement [15]. StreamCorrect adds learned correction and feeds selected corrections into subsequent recognition [7]. RevTrace retains that selection and feedback behavior in its controlled native adapter; its contribution concerns the asynchronous execution and consumption around the feedback boundary. It does not introduce a better acoustic model or train a new corrector.

SSBD coordinates old-output draft reuse with display-only suffix masking [5]. It revalidates a draft under updated input and includes an unbiased configuration. RevTrace follows complete calls that actually executed to establish their eligibility under newly selected feedback state, then checks the dependency witness for the Consumer's whole current view. PredGen provides a close precedent for speculation while speech input evolves across recognition, language generation, and speech synthesis [6]. Our comparison holds the complete selected feedback task fixed across the retention policies. The specific reuse object, protected selection, and whole-view transition define the composition evaluated here. RelayS2S uses a learned verifier to commit a fast-path prefix that constrains subsequent slow-path generation [16]. Our native calls instead confirm their complete inputs and retain the original correction and feedback update. Work on ASR alignment drift also establishes the importance and cost of audio position state [17]; our contract includes that state rather than treating text equality as sufficient.

*c) Execution, authorization, and recovery:* Retire studies versioned execution and bounded physical exit for interruptible agents [18]. RevTrace likewise distinguishes logical progress from an executor's actual release, while deriving the tested retention and coverage decisions from the feedback contract. Our simple covered-exit implementation is execution support, without a claim to a new general cancellation algorithm. A parent-version cancellation control is not a reproduction of the complete Retire system.

Optimistic and multiversion concurrency control separate validation and commit from concurrent execution [19], [20]. Write-ahead recovery persists state changes and identities [9], and stream-processing systems coordinate recoverable state [21]. We use these established principles at the ASR feedback boundary. They supply machinery for ordering and durability; the adapter still has to specify which state constitutes a valid continuation and which complete view the Consumer may retain.

## VII. Discussion

*a) Scope of the guarantee:* The continuation witness assumes a sufficient adapter state and faithful execution of frozen inputs. A new adapter must account for every mutable field that can affect later inference, selection, or feedback. A hidden KV cache, alignment cursor, or external tool state can invalidate a token-only argument. Conservative rejection remains allowed when a sufficient witness is absent. The guarantee concerns dependency validity and an individual compatible application; it supplies neither transcription correctness nor monotone text for an entire session.

*b) Progress, content, and capacity:* Processing more audio can leave the cumulative transcript unchanged. Frontier credit, actual text additions and revisions, eventual selected text agreement, and ground-truth accuracy answer different questions. We report them with their original denominators. Preserving compatible progress also cannot remove all input synchronization, queueing, or neural work. The negative c12 witness and the native two-replica comparison identify concrete limits of this implementation rather than a universal capacity envelope. The independent native confirmation establishes a single-chain service opportunity with the original selected task; it does not turn that result into saturated multi-source capacity.

*c) Deployment and validation:* The continuous-original reference exercises the native cut across real buffer and context changes; another adapter must establish its own state sufficiency. Active-dependency collection bounds protocol entries relative to live references, separately from cumulative text, current state, and total memory. Owner/worker recovery assumes the surviving services of Section II. Replaying old facts restores access to that prefix; the measured cold Owner restarts still miss original deadlines for new work. Broader ownership transfer requires separate validation. Finite specifications, source controls, controlled faults, actual GPU service, and retrospective checks provide distinct evidence.

*d) Eligibility and resource admission:* A valid input witness determines whether an owned call may be used. It does not determine whether starting another conditional call is worthwhile. Limiting new speculation can leave existing supported whole views intact. The long stress study motivates this separate resource decision. In a bounded two-client probe, new speculative calls precede another client's ready necessary call in both repeats. The later client has no additional C8 misses, and its C99 penalty fails the predeclared repeat-consistency gate. No admission controller or general native-session fairness is evaluated.

## VIII. Conclusion

RevTrace separates complete-input eligibility, original selection authority, and whole-view support while trusted feedback advances. Service experiments expose a coordination gap: first delivery can meet its deadline, yet a compatible update can withdraw that progress. The joint design protects the supported ahead view. Native full calls can retain original correction work after their inputs are confirmed even when their own next predictions are wrong. Durable facts and physical identities carry this distinction through supported process failures. The resulting composition connects dependency reuse to actual continued consumption under a fixed feedback task, with measured timing, content, recovery, and capacity boundaries. Extending recovery to Supervisor failure remains a separate direction.

## AI Tool Usage

ChatGPT/Codex assisted literature discovery, drafting and revision of all manuscript sections and the refinement argument, runtime extensions and experiment orchestration, and validation/analysis code producing the figures and tables from recorded measurements. The artifact records execution sources, frozen configurations, controls, and result provenance, separating actual service measurements from component tests and retrospective analyses. Constructed streams concatenate recorded AISHELL-1 PCM. The authors retain responsibility for the submission.

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